

Implications of Carbon Tax and Energy Efficiency Improvement of CO₂ Emission Reduction Targets on Energy and Environment in Thailand¹

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Abstract

This study examines energy system development and its associated greenhouse gas and local air pollutant emissions under base case and three different emissions reduction targets in Thailand during 2000-2035. Base case results show that fossil fuels would continue to dominate the total energy supply and the corresponding CO₂ emissions would increase by more than four-folds between years 2000 and 2035. Power-, transportation- and industry- sector together would contribute 93% of total CO₂ emissions by 2035. With limited availability of economically viable new and renewable energy sources and indigenous fossil fuels, the country's energy import dependency under the base case would reach 77% in 2035 from 50% in 2000.

With ER targets set from 2013, the results of the study show a fall in coal share and an increase in natural gas share in the total primary energy-mix during 2013-2035. However, the new and renewable energy sources would play a limited role in primary energy-mix. The power sector would play a higher role in reducing CO₂ emissions than the other sectors especially at high ER targets. Under ER15, up to 41% of SO₂ and 10% of NO_x emissions could be reduced as co-benefits. New technologies such as carbon capture and storage, and solar for electricity generation; and hydrogen based fuel cell- and hybrid- vehicles for road transportation would become cost-effective under ER targets. The incremental CO₂ mitigation cost would vary from 36 US\$ per t-CO₂ under ER target of 5% to 49 US\$ per t-CO₂ under ER target of 15% during 2013-2035.

Keywords: Post-Kyoto CO₂ emission reduction target; Energy system development; Thailand AIM/Enduse model

6.1 Introduction

Developing countries are not legally required by the Kyoto Protocol (KP) of the United Nations Framework Convention on Climate Change (UNFCCC) to reduce greenhouse gas (GHG) emissions. However, the first commitment period of the KP runs only up to 2012 and the global community has already begun discussing the future climate regime beyond 2012 (i.e., post-Kyoto) that involves active participation from developing countries (DCs). In anticipation of growing national and international concerns for mitigation of GHG emissions and increasing pressure on DCs to participate in the post-Kyoto climate regime, it is important to identify sustainable energy pathways that lead to low carbon economy in DCs in the future. This requires identification of various options for mainstreaming climate concerns developmental policies and strategies of individual DCs. Thus, it is of interest for policy makers to analyze the implications of post-Kyoto CO₂ emissions reduction (ER) targets on energy and environment in developing countries. A number of studies have systematically analyzed several options of emission reduction requirements and time of participation in the international climate regime (Jacoby et al., 1999; Berk and den Elzen, 2001; Criqui et al., 2003; den Elzen et al., 2005; Groenenberg et al., 2004; Höhne et al., 2002; Kawase et al., 2006; WBGU, 2003; WWF, 2005).

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However, most of these studies are concentrated at the global or Annex I² level of analysis and there are limited studies on implications of ER on energy system in DCs. A recent study by IGES (2005) highlighted concerns, interests and priorities from Asian perspectives on climate regime beyond 2012 but limited reference were made on ER targets and their implications on energy systems in developing countries. In this study, we first analyze energy system development and its implications on environment in the context of Thailand without any constraint on ER during 2000-2035. Then an assessment of post-Kyoto ER targets on energy systems and environment is made over the study period starting from 2013 through 2035. In particular, this study assesses implications of CO₂ ER targets on changes in primary fuel-mix, sector-wise energy-mix, use of renewable energy technologies, sector-wise GHG emissions and other local pollutants, and energy system costs for the post-Kyoto period of 2013 up to 2035. To analyze these issues in detail, an energy system optimization model is developed for Thailand in the AIM/Enduse model framework which is a member of the Asia-Pacific Integrated Assessment Model (AIM) family.³ The methodology used is essentially an end-use approach that interrelates the energy service demand with the population, gross domestic product (GDP), energy price, technological change, etc.

The chapter is divided into six sections. Section 6.2 describes the methodology used in the study. Section 6.3 provides the scenario description including a brief description of data used and their sources. Analyses of results and key findings are presented in Section 6.4. The final section presents key findings and final remarks.

6.2 Methodology

The AIM/Enduse model of Thailand is broadly classified into two main components: (i) energy supply and conversion and (ii) service demand. The energy supply and conversion component represents energy extraction, imports and conversion of primary energy to secondary energy. In this component, coal mining, natural gas extraction, refining of crude oil and power generation is considered. For power generation, altogether twenty-seven existing and new technologies option are considered. Out of the new technology options considered, five technologies are coal- and natural- gas based carbon capture and storage (CCS) technologies. A study by NEPO (1999) is used as the basis for classification of sectors and sub-sectors of the service demand. The Thai economy is divided into five main sectors; namely, agriculture, commercial, industry, residential and transport. The industry sector has been subdivided into cement, steel, sugar, paper, chemicals, food, equipment, textile and others. The residential sector has been divided into urban and rural categories while the transport sector is divided into passenger and freight categories. The end-use service demand types and the activity parameters considered in AIM/Enduse model for Thailand are summarized in Appendix 6.A. These are exogenous inputs to the model that are used in estimating future service demands. Altogether 292 existing and candidate technology options are considered in the study for meeting these end-use service demands. Figure 6.1 shows the general structure of the AIM/Enduse modeling framework for Thailand.

² Annex I Parties include the industrialized countries that were members of the OECD (Organization for Economic Co-operation and Development) in 1992, plus countries with economies in transition (the EIT Parties), including the Russian Federation, the Baltic States, and several Central and Eastern European States.

³ The Asia-Pacific Integrated Model (AIM) is developed by the National Institute for Environmental studies (NIES), Japan. This model is demand driven based on a partial equilibrium framework that uses bottom-up technologies, applied for estimating and analyzing future sector-wise energy and emissions in Asian countries. See Kainuma et al. (2003) for details on AIM.

Figure 6.1 here

The linear programming framework of the model comprises an objective function to minimize total energy system cost year by year subject to a number of constraints including those on service demand, energy resource availability, existing device stock, maximum allowable quantity of devices and emissions. The total cost comprises of annualized fixed cost of recruited devices during a year, variable operating cost (i.e., operation and maintenance cost of devices, and fuel cost), cost of installing removal devices (e.g., flue gas desulfurizers for pulverized coal fired power plants etc.) and cost of emissions taxes (such as carbon and energy taxes). The formulation also provides functions to consider the existing device quantities in the starting year of the planning horizon and to calculate the retirement of the devices at the end of its life time.

6.3 Scenarios Description

Four scenarios are considered in this study; i.e., a base case and three alternative cases each of which consider different ER targets. The base year is considered as 2000 and the end year of the planning horizon is 2035. The data on economic and demographic parameters, service demand, technology characteristics, energy price, domestic resources availability, emission factors for CO₂, SO₂ and NO_x, appliance stock in the start year (i.e., 2000) and the maximum share of each technology type are collected from several sources and these sources as described in [Appendix 6.B](#). The data obtained from the original source are processed to meet the input requirements of AIM/Enduse framework.

The base case scenario is defined as the business-as-usual, i.e., continuation of current economic, demographic and energy sector trends and policies, and there is no mitigation policy. [Appendix 6.C](#) summarizes the maximum availability of domestic fossil-fuel resources (coal, oil and natural gas) during 2000-2035 under the base case. However, no limit is imposed on imports of these fossil-fuel resources. New and renewable energy (NRE) sources include solar, wind, hydro, geothermal, firewood, charcoal, municipal solid waste (MSW), biogas, agricultural residues, plantation based biomass and biofuels (i.e., ethanol and biodiesel). The maximum potential of solar and wind for power generation are considered as 5,000 MW and 1,100 MW, respectively ([Greacen, 2005](#)). Likewise, the maximum exploitable potential of agricultural residues considered is 14,112 ktoe which comprise of sugar cane residues (i.e., bagasse and top and leaves), paddy husk, corncob and others ([Santisirisomboon et. al., 2003](#); [Prasertsan et. al., 2006](#) and [DEDE, 2006e](#)). Considering the land area availability, the potential of plantation based biomass considered is 7,500 ktoe ([Sajjakulnukit and Verapong, 2003](#) and [Santisirisomboon et. al., 2003](#)). The plantation based biomass, considered only for power generation, is produced on sustainable basis and therefore the corresponding CO₂ emissions is assumed zero. Following the government's plan, the potential of biofuels considered are 3 million liters per day of ethanol and 8.5 million liters per day of biodiesel ([DEDE, 2006d](#)). The discount rate considered is 10%. All costs are expressed in constant 1995 US\$.

In the three emission reduction cases, targets for CO₂ emissions reduction are set at 5%, 10% and 15% of the base case emission considered during the post-Kyoto period of 2013-2035. Hereafter, these three cases are referred to as ER5, ER10 and ER15.

6.4 Results and discussions

In this section, the results of the analysis of energy pathways and their associated environmental and cost implications in Thailand are presented under the base case and three different targets for ER.

6.4.1 Base case

Energy implications

Increasing use of fossil fuels and dominant shares of transport and industry sectors in total energy systems of the country characterizes the base case. The present study shows that in this scenario, the total primary energy supply (TPES) would increase by more than three-folds from 77 Mtoe in 2000 to 285 Mtoe in 2035 and the share of fossil fuels (coal, oil and natural gas) in TPES would increase from 81% in 2000 to 87% in 2035 (Figure 6.2). Although the oil demand would increase in absolute terms, at an average annual growth rate (AAGR) of 2.4% during 2000-2035, its share in TPES would fall from 45% in 2000 to 34% in 2035 due to partial substitution of gasoline and diesel by biofuels and natural gas used in road transportation. On the other hand, the share of coal would increase from 11% in 2000 to 28% in 2035. Coal use would be essentially restricted to industrial applications (cement and steel production) and electricity generation. The share of natural gas would remain about the same, at 25% in 2000 and 2035. In contrast to fossil fuels, the share of NRE sources (i.e., biomass, hydro and new renewables) in TPES would fall from 19% in 2000 to about 13% in 2035. The share of biomass in total NRE sources would fall from 95% in 2000 to about 85% in 2035. However, the share of hydro and new renewables (i.e., wind, solar, geothermal, MSW, biogas, ethanol and biodiesel) together in total NRE sources would increase to 15% by 2035 as compared to only about 5% in 2000.

The country's energy import dependency (EID) – defined as net energy import as a percentage of TPES – would increase from 50% in 2000 to 77% in 2035 showing increasing vulnerability to energy supply uncertainties related to external sources. The country's high fossil fuel dependency is due to limited indigenous availability of fossil fuels and limited potential of economically viable new and renewable energy sources such as biomass solar, wind, geothermal and hydro.⁴

Figures 6.2 and 6.3 about here

As seen from Figure 6.3, the total final energy demand (FED) would increase from 49 Mtoe in 2000 to 215 Mtoe in 2035, an increase of more than four-folds. The AAGRs of sector-wise FED over the study period are: 2.3% for agriculture, 6.1% for commercial, 5.0% for industry, 1.6% for residential and 4.1% for transportation. The transport- and industry- sector together accounted for almost 74% of the total FED in 2000. As the country's economy is estimated to grow, the share of these two sectors combined in the total FED would further increase to 81% by 2035.

CO₂ and local air pollutant emissions

⁴ According to DEDE (2004a), Thailand's proven reserves of lignite and natural gas stand at 2,780 Mt and 354 billion m³ respectively. Assuming the current rate of production and no new reserves discovery in the future, indigenous natural gas resource would be exhausted by 2020 while lignite would last for next 140 years or so.

Country's energy related CO₂ emissions would increase at an AAGR of 4.1%, i.e., from 158 Mt in 2000 to 676 Mt in 2035. As seen from [Figure 6.4](#), power-, industry- and transport- sector together would contribute about 93% of total CO₂ emissions during 2000-2035. In 2035, the industry sector would contribute 40% of the total CO₂ emissions, followed by transport- (29%) and power- (25%) sector. In the same year, agriculture-, residential- and commercial- sector together would contribute the remaining 7% of the total CO₂ emissions. The low share of power sector as compared to industry- and transport- sector in total CO₂ emissions in 2035 is due to the adoption of carbon capture and storage (CCS) based electricity generation technologies. Likewise, the results of the study find that local air pollutant emissions would also increase in the country during the study period. The SO₂ emission would increase by almost five-folds (from 899 kt in 2000 to 5,604 kt by 2035) while the NO_x emission would increase by four-folds (926 kt in 2000 to 3,413 kt in 2035) during 2000-2035 ([Figure 6.5](#)).

 Figures 6.4 and 6.5 about here

In terms of per capita CO₂ emissions in the country, it would more than triple (i.e., 2.7 tons per person in 2000 to about 9.2 tons per person in 2035) over the next 35 years. However, the estimated 2035 figure of emission per capita in Thailand would still be comparable to year 2002 emission per capita figures of some of the industrialized countries (e.g., 10.2 tons per person in Germany, 9.5 tons per person in Japan and South Korea, and 8.9 tons per person in United Kingdom) and it would be almost twice the projected 2030 level of the world as a whole (i.e., 4.7 tons per person) ([IEA, 2004a,b](#); [World Bank, 2006](#);). Based on the post-Kyoto per capita emission proposals put forward in recent literature (see e.g., [Sijm et al., 2001](#); [Gupta, 2003](#); [Groenenberg et al., 2004](#); [Bode, 2004](#); [Blok et al., 2005](#)), Thailand would fall in the category of countries that would require GHG emissions reduction commitment. In contrast, the overall CO₂ emission intensity in the country - defined as kg of CO₂ emission per US\$ of GDP at 1995 prices using the market exchange rate - is estimated to decline by 22% during 2000-2035, i.e., from 0.98 in 2000 to about 0.72 in 2035. However, Thailand's CO₂ emission intensity would still be much higher than the corresponding projected figures in 2030 of 0.45 and 0.30 for the world and OECD as a whole, respectively.

6.4.2 CO₂ emissions reduction target case

Effects on primary energy-mix and final energy demand

With ER targets set from 2013, the results indicate substantial changes in the primary fuel-mix during 2013-2035. As seen from [Figure 6.6](#), coal share in TPES would fall during the period: It would reduce by 6% under ER5 case and by 12% under ER10 and ER15 cases as compared to the base case. The reason for only 12% fall in coal share under ER15 is due to increasing use of CCS based electricity generation (CCS based power generation primarily uses coal and natural gas) with increasing ER targets. On the other hand, natural gas share in TPES would increase steadily by 2%, 7% and 10% under ER5, ER10 and ER15 cases respectively. The reason for rise in the share of natural gas in TPES over the study is its increasing use in electricity generation and industries. Likewise, the share of others (biomass, hydro and new renewable energy sources) share in TPES would increase steadily by 1%, 4% and 10% under ER5, ER10 and ER15 cases respectively. In contrast, changes in oil share under all ER cases are rather less significant. The share of oil in TPES over the study period would increase by

0.3% and 0.6% under ER5 and ER10 cases respectively, while it would fall by 2% under ER15 case as compared to base case. Most of the additional oil requirements to meet growing travel demand would be met by increasing use of biofuels (ethanol and biodiesel). While no biofuels is used in road transportation in the past, its share in TPES for transportation would reach 6% (3.3 Mtoe) under ER15 by 2035.

Figures 6.6 and 6.7 about here

The study shows that new and renewable energy (NRE) sources would play a limited role in primary energy-mix under ER targets. Among the NRE sources, the use of biomass (excluding biofuels), wind and solar is found to increase with increasing ER emission targets (Figure 6.7). Solar power (PV and thermal) would be economical only under ER10 and ER15 cases while biomass (excluding biofuels) and wind power would be attractive under all ER cases. However, the increase in the use of hydro, geothermal, biogas and MSW resources with increasing ER targets would remain limited under all ER cases. Overall, the total share of NRE sources is estimated to reach 18% under ER15 case as compared to base case value of 17.5% during 2013-2035.

With increasing ER targets, the FED would also reduce in increasing order during 2013-2035 by 1.8% (61 Mtoe), 2.4% (82 Mtoe) and 3.8% (130 Mtoe) under ER5, ER10 and ER15 cases respectively. As seen from Figure 6.8, the transport sector would contribute the most in the reduction of total FED at 3.1% under ER5 to 6.7% under ER15 during 2013-2035, followed by industry sector at 1.6% under ER5 to 3.2% under ER15 and residential sector 0.2% under ER5 to 0.5% under ER15. The reduction of the total FED from agriculture and commercial sectors are however, negligible.

Figure 6.8 about here

Changes in sectoral CO₂ emissions

With ER targets, most of the CO₂ emission reduction would take place in the power, industry and transportation sectors as these sectors would consume more than 90% of TPES in the country. As seen from Figure 6.9, the power sector alone would contribute most of CO₂ emissions reduction varying from 44% under ER5 to as high as 58% under ER15 during 2013-2035. This is because the power sector provides the highest flexibility to fuel switching (coal to natural gas) and the adoption of new technologies (integrated gasification combined cycle (IGCC) with CCS) under ER targets. The results demonstrate that the power sector would play a higher role in reducing CO₂ emissions than the other sectors especially at high ER targets. The industry sector is the second largest after the power sector, accounting for more than 33% and 22% of total CO₂ emission reduction under ER5 and ER15 cases respectively during 2013-2035. The reduction in this sector would come primarily from fuel switch, i.e., from coal to natural gas. The transport sector is the third largest sector after power- and industry- sectors, accounting for 23% and 20% of CO₂ emission reduction under ER5 and ER15 cases respectively during 2013-2035. The reduction of CO₂ emission in transport sector would come mainly from the use of more efficient vehicle technologies.

Figure 6.9 here

Effects on changes in energy-mix and technologies for electricity generation

Over the past 15 years, the total electricity generation in Thailand grew much faster (8% per year) than its GDP (4% per year) (NIDA, 2006; NESDB, 2006). Unlike in the past, the study shows that electricity generation in the base case would increase at much lower rate (4.5% per year) during 2000-2035.

As shown in Table 6.1, base case total electricity generation is estimated to increase from 93 TWh in 2000 to about 433 TWh in 2035. With ER targets, the total electricity generation in 2035 would be at slightly lower under ER5 case (433 TWh), while it would be slightly higher at 445 TWh under ER10 and ER15 cases as compared to base case. The reason for the lower total electricity generation under ER5 as compared to the base case is that more efficient and advance electricity generation technologies (such as advanced natural gas combined cycle) would be used, while the reason for higher total electricity generation under ER10 and ER15 cases is due to increasing use of CCS technologies (which are less efficient than advance combined cycle) with increasing ER targets. Figure 6.10 shows the energy-mix in electricity generation under the base case and three ER target cases during 2013-2035. Compared to the base case, natural gas share in total energy requirements in the power sector would increase by 16% (i.e., 53 Mtoe), while coal share would fall by 12% (i.e., 46 Mtoe) under ER15 during 2013-2035. The fall in coal share is due to a shift of existing coal-based thermal plants to more efficient new pressurized fluidized bed combustion (PFBC) and IGCC plants with CCS technologies. The results of this study indicate that under ER15, electricity generation from coal- and natural gas- plants with CCS technology would require about 23% of total coal requirements and 5% of total natural gas requirements during 2013-2035. With ER15, the study shows that within NRE sources for electricity generation, the share of new renewables, biomass and hydro would increase by 22% (i.e., 12 Mtoe), 4% (i.e., 8 Mtoe) and 1% (i.e., 0.3 Mtoe) respectively as compared to base case during 2013-2035.

Table 6.11 about here

In terms of technology mix for electricity generation, the study shows substantial changes under the ER targets. As seen from Table 6.1, integrated gasification combined cycle (IGCC) with CCS technology and advanced natural gas combined cycle plants would become economically viable options for electricity generation under ER targets. In 2035, about 11% of total electricity generation would be based on CCS based electricity generation technologies under the base case and the share is estimated to increase up to 17%, 25% and 35% under ER5, ER10 and ER15 cases respectively. Although relatively small, the share of advance natural gas combined cycle with CCS technology plants would increase from 3% under ER10 to about 7% under ER15 in 2035. In the case of NRE based power generation technologies, their share would increase from 9% in 2000 to about 29% in 2035 under ER15. Excluding biomass based electricity generation, the share of hydro and new renewables (i.e., wind, solar, geothermal, MSW and biogas) based power generation technologies would increase from 9% in 2000 to about 12% and 14% in 2035 under ER10 and ER15 cases respectively. More than two-thirds of the total NRE based electricity generation in the country would come from biomass and hydro under ER targets in 2035. However, the share of biomass and hydro based electricity

generation in total NRE based electricity generation would decline with increasing ER targets varying from 84% under ER5 to 77% under ER15 in 2035 due to increasing share of solar. Among the new renewable technologies, solar would become economically viable at ER10 and ER15 cases in 2035 accounting for 9% and 10% of the total NRE based electricity generation under ER10 and ER15 respectively, while geothermal, MSW, biogas and wind would account for 5%, 5%, 2% and 2% respectively in 2035.

Figure 6.10 about here

To strengthen energy security and sustainability, the Thai government has set 2011 as the target date for 8% of the nation's total energy to be generated from renewable energy (biomass and new renewables excluding hydro) sources. A key component of this renewable strategy is the Renewable Portfolio Standard (RPS) that calls for new power plants to use renewable energy with 5% of their total capacities by 2011 (EGAT, 2004). Renewable sources excluding hydro accounted for only about 1% of electricity generated in 2004. With ER targets, the share of NRE (excluding electricity generation from hydro) in total electricity generation would increase up to 21% and 22% under ER5 and ER15 cases by 2035. However, the share of new renewables in country's total electricity generation would only be about 4% and 7% under ER5 and ER15 in 2035. The low share of new renewables is due to limitations in the potential availability of geothermal and wind power, and high initial cost of solar.

Effects on road transport fuels and technologies

With ER targets, the present study finds that the conventional vehicle technologies would be replaced by hybrid and fuel cell vehicle technologies in road transportation. As seen from Figure 6.11, the share of hybrid vehicle technology would increase from 49% under the base case to 55% under ER15 case in 2035. Likewise, the share of fuel cell vehicle technology would increase from about 20% under the base case to 22% under ER15 in 2035. The new and advance vehicle technologies could reduce the total road transportation energy requirements by 4% under ER5 and by 8% under ER15.

In May 2005, the Thai government approved a new energy strategic plan. One of the targets set under the plan is to reduce oil consumption by 25% from the road transport sector by 2009 by promoting biofuels (i.e., ethanol and biodiesel) and natural gas use in motor vehicles. This study finds that biofuels and natural gas together could meet only about 6.5% of total energy requirements in road transportation in 2009. However, the share of alternative fuels could increase up to 24% (i.e., 10% from hydrogen, 6% from biofuels and 8% from natural gas) by 2035 in the base case. On the contrary, there would be a reduction in natural gas requirements in road transportation under ER targets. This is due to the switch of low operating efficiency natural gas vehicles by more efficient new and hybrid vehicles. Indeed, only about 6% of total energy requirements in road transportation would come from natural gas while about 7% each would come from hydrogen and biofuels under ER15 during 2013-2035.

Figure 6.11 about here

Co-benefits under CO₂ emission targets

The CO₂ ER targets results in reducing SO₂ and NO_x emissions as co-benefits. [Figure 6.12](#) shows the percentage reduction of SO₂ and NO_x emissions under each ER target cases as compared to base case during 2013-2035. As seen from the figure, ER targets would be more effective in reducing SO₂ emissions than NO_x emissions. This is because relatively high sulfur content coal would be substituted by natural gas under ER cases resulting in higher decrease in SO₂ emissions than NO_x emissions in percentage terms. Compared to base case, the percentage reduction in total SO₂ emission during 2013-2035 would be 13% (12.5 Mt), 31% (30 Mt) and 41% (40 Mt) under ER5, ER10 and ER15 cases respectively. Likewise, the percentage reduction in total NO_x emission during 2013-2035 would be 2% (1 Mt), 5% (3 Mt) and 10% (6 Mt) under ER5, ER10 and ER15 cases respectively. The reduction in NO_x emissions under ER targets could be attributed partial substitution of coal and oil by natural gas in the power and industry sector and increasing use of natural gas vehicles in road transportation.

As seen from [Figure 6.13](#), the industry and power sectors together would account for about 97% and 94% of total NO_x emission reduction during 2013-2035 under ER5 and ER10 respectively, while the transport sector would account for 38% of total NO_x emission reduction under ER15. Likewise, power and industry sectors together would account for 84% and 86% of total SO₂ emission reduction during 2013-2035 under ER10 and ER15 cases. However, under ER5, the results show that there would be an increase of 54% in SO₂ emission from power sector due to more use of lignite for electricity generation.

 Figures 6.12 and 6.13 about here

The cost of CO₂ emission abatement

The incremental abatement cost (IAC)⁵ of mitigating CO₂ emission is presented in [Figure 6.14](#). As seen from the figure, the CO₂ mitigation cost would increase by 63 US\$ per t-CO₂, 66 US\$ per t-CO₂ and 125 US\$ per t-CO₂ under ER5, ER10 and ER15 cases respectively in reducing the corresponding CO₂ emissions of 459, 965 and 1,486 Mt-CO₂ during 2013-2035. This gives an indication on how some of the CO₂ mitigation projects in Thailand could be attractive for funding under the Clean Development Mechanism (CDM) as stipulated in the KP of the UNFCCC.⁶ The prices estimated in the study are much higher than the current price of certified emission reductions (CERs) which is in the range of 3 to 6 US\$/t-CO₂ ([Pacudan and Lee, 2006](#)). However, the IAC figures estimated for Thailand is comparable with figures of the marginal

⁵ The incremental abatement cost (IAC) of CO₂ emissions is calculated as follows:

$$IAC = TC^e - TC^0 / \left\{ \sum_{t=1}^T (E_t^0 - E_t^e) / (1+r)^t \right\}$$

where, TC^e = present value of total cost corresponds to the least cost generation expansion plan with emission constraints, TC^0 = present value of total cost corresponds to the least cost generation expansion plan without emission constraints, E_t^0 = pollutant emission in year t corresponds to the least cost generation expansion plan without emission constraints, E_t^e = pollutant emission in year t corresponds to the least cost generation expansion plan with emission constraints, r = discount rate, T = planning horizon.

⁶ Under the Clean Development Mechanism (CDM) of the Kyoto Protocol, certain GHG mitigation projects in countries which have not committed for GHGs reduction could be developed/financed by parties from Annex-I countries. These countries could, in return, get certified emission reductions (CERs) which could be used to meet a part of their GHG mitigation obligation under the UNFCCC ([UNFCCC, 1998](#)).

abatement costs at the levels of abatement required in selected industrialized countries and the regions of the world. For example, it is 51 US\$/t-CO₂ for USA, 159 US\$/t-CO₂ for Japan, 74 US\$/t-CO₂ for European Union as a whole and on average 64 US\$/t-CO₂ for other OECD countries (Shrestha and Marpaung, 2002).

Figure 6.14 here

6.5 Conclusions and final remarks

Thailand is a heavily fossil fuel intensive economy and is highly dependent on imported energy. Energy security is high on Thai government's agenda. Indeed, more than half of the total primary energy requirement was met through import in 2000 and this study shows that the energy import dependency of the country would reach 77% by 2035. Thailand currently contributes only about 0.7% of the global energy related CO₂ emission. However, if the current trend continues without any mitigation measure, CO₂ emission from the country would increase by more than four-folds by 2035 as compared to the 2000 value. The power sector, which is predominantly fossil-fuel based, would contribute the most in total CO₂ emissions, followed by the transport and industry sectors. These three sectors together would contribute more than 93% of the total CO₂ emissions from the country during 2013-2035.

There are significant implications of introducing post-Kyoto CO₂ emissions reduction targets for the structure of energy system as well as CO₂ and other local air pollutant emissions. With increasing ER targets, coal share in TPES would fall while natural gas and NRE shares would increase during 2013-2035. However, the fall in coal share in TPES (i.e., 12%) would remain the same under ER10 and ER15 cases due to increasing use of coal based CCS technologies for electricity generation under ER15. Indeed, the share of coal- and natural gas- based CCS technologies would increase by 17%, 25% and 35% under ER5, ER10 and ER15 cases respectively as compared to only about 11% under the base case during 2013-2035. Among the NRE sources, the use of biomass, wind and solar is found to increase with increasing ER emission targets during 2013-2035. While no biofuels is used in road transportation in the past, its share in TPES for road transportation would reach 6% (i.e., 3.3 Mtoe) under ER15 by 2035.

The power sector would play a higher role in reducing CO₂ emissions than the other sectors especially at high ER targets varying from 44% under ER5 to as high as 58% under ER15 during 2013-2035. Industry- and transport- sector is the second and the third largest after the power sector accounting for 22% and 20% of total CO₂ emission reduction respectively under ER15 by 2035. ER targets would be more effective in reducing SO₂ emissions than NO_x emissions due to the substitution of high sulfur content lignite by natural gas resulting in higher decrease in SO₂ emissions than NO_x emissions.

The CO₂ mitigation cost is estimated to vary from 36 US\$ per t-CO₂ under ER5 case to 49 US\$ per t-CO₂ under ER15 case during 2013-2035.

AIM/Enduse model for Thailand is a static year by year optimization model. Thus, results presented in this study could be different from end-use models that consider dynamic optimization over the planning horizon. In addition, effects of trade and structural repercussions from changes in energy prices and costs throughout the economy are not considered due to the technology-specific bottom-up partial equilibrium approach of AIM/Enduse model. Furthermore, some of the features of the electricity generation sector such

as lumpy nature of investments, discrete nature of power generation capacities and variations in daily load curve are not considered in the selection of power generation plants. Due to data limitations, options of energy from nuclear based technologies in the electricity generation in Thailand are not included in this study. In a subsequent paper, we intend to consider the role of nuclear technologies in the energy system development and the environmental management of Thailand.

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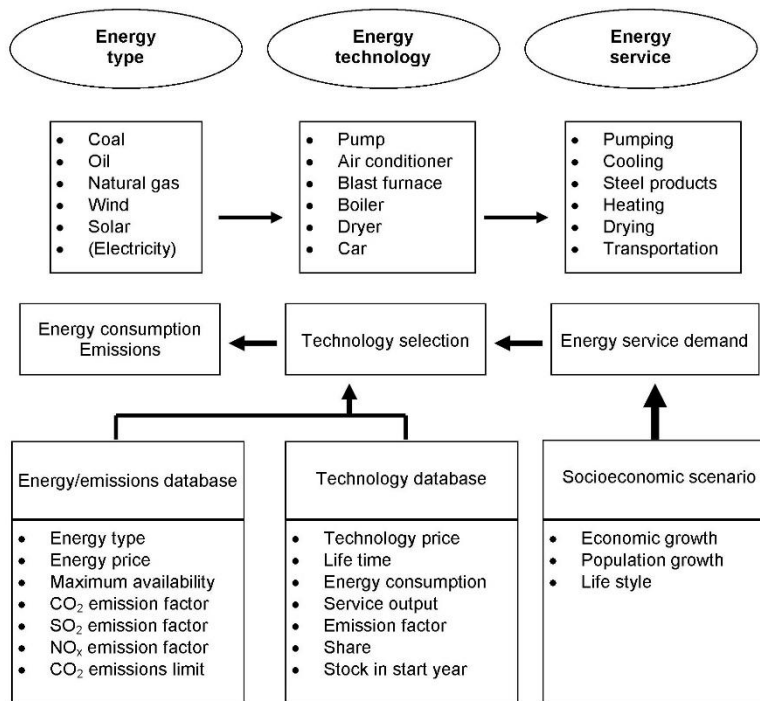
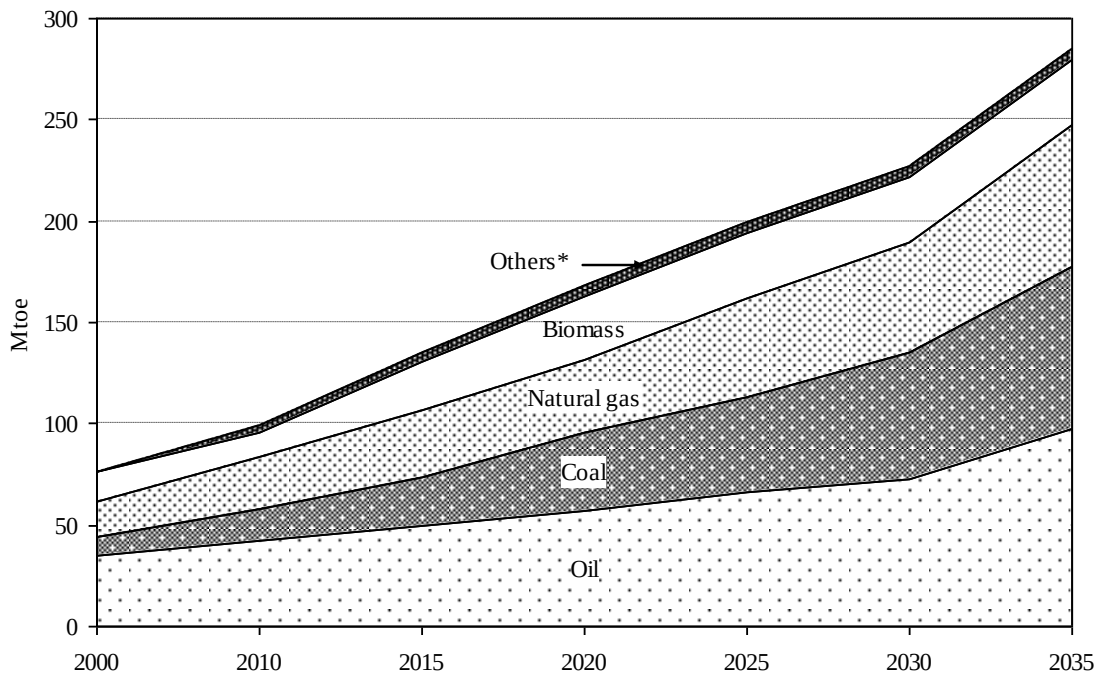


Fig. 6.1: Structure of the Asia-Pacific integrated assessment model (AIM)/Enduse model.



Note: * Others includes hydro, solar, wind and geothermal.

Fig. 6.2: Primary energy supply mix under the base case 2000-2035.

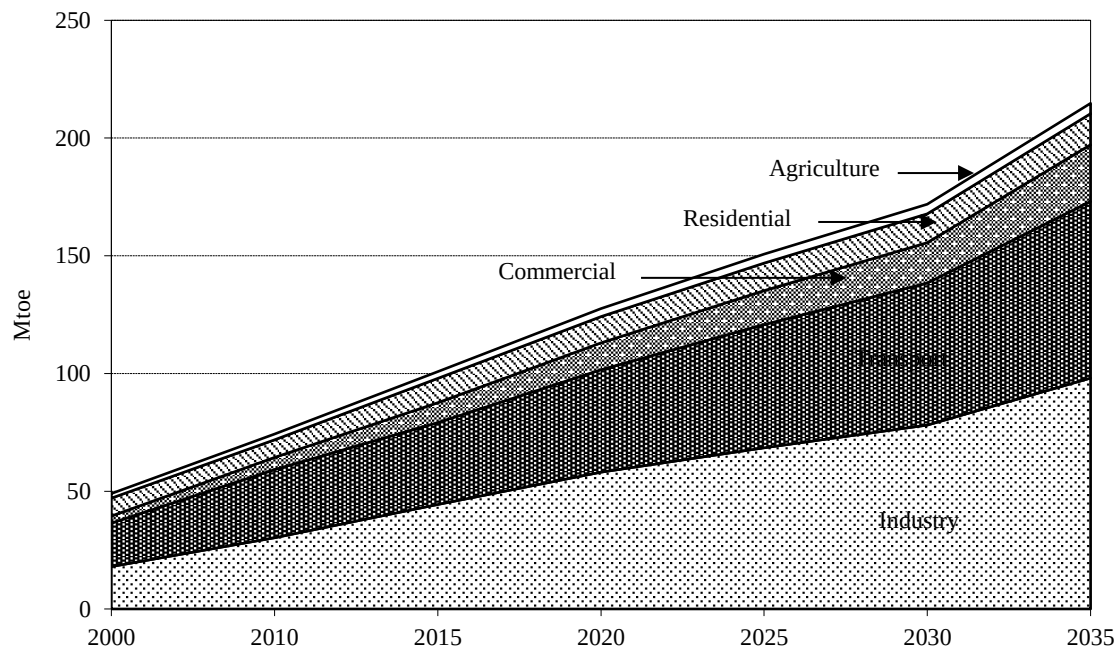
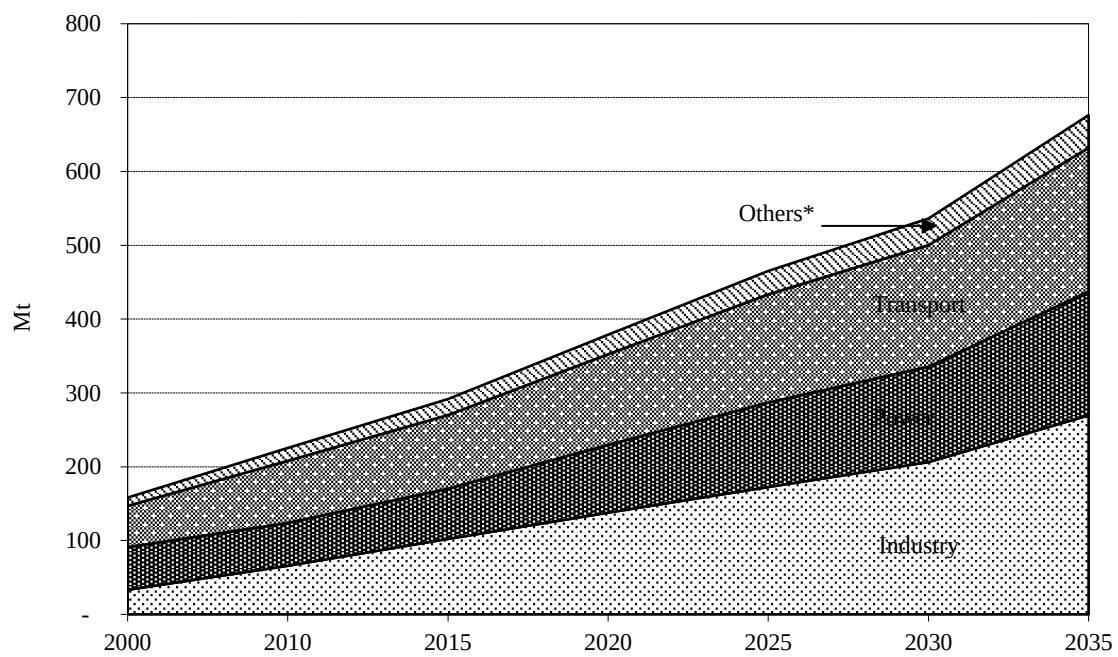
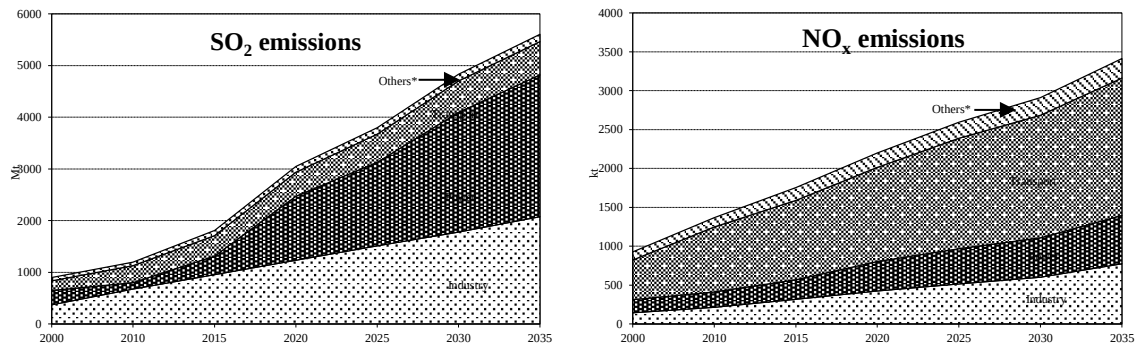


Fig.

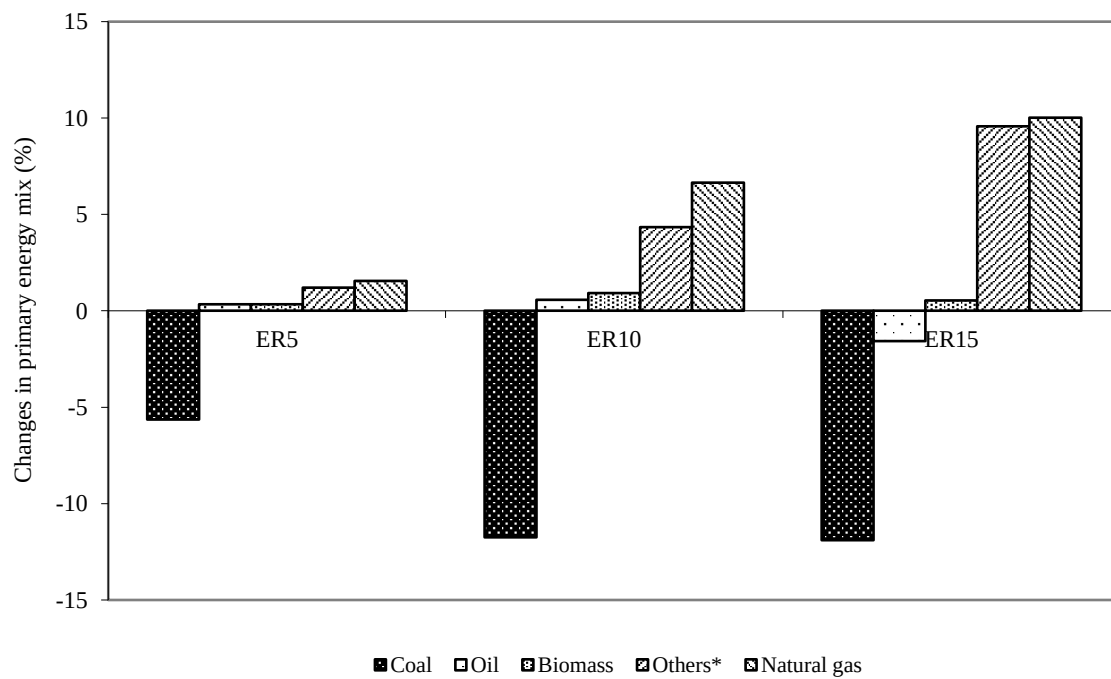
6.3. Final energy demand by sector under base case during 2000-2035.



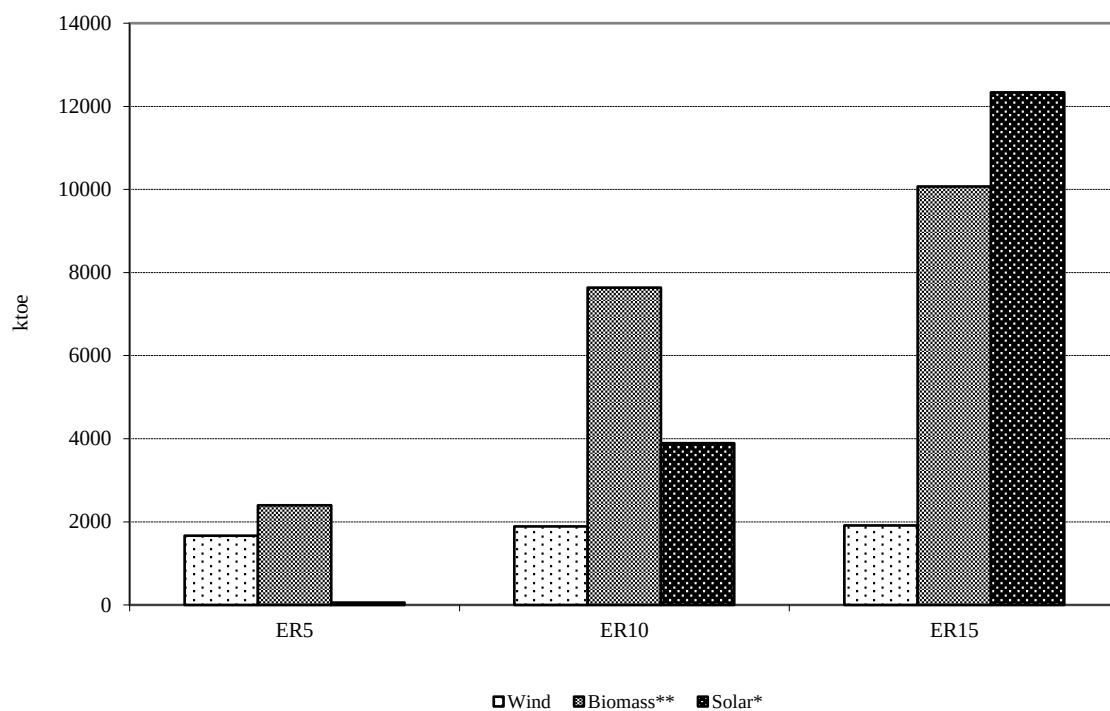
Note: * Others include agriculture, commercial and residential sectors.
Fig. 6.4. CO₂ emissions by sector under base case during 2000-2035.



Note: * Others include agriculture, commercial and residential sectors.
Fig. 6.5. SO₂ and NO_x emissions by sector under base case during 2000-2035.



Note: * Others includes hydro, solar, wind and geothermal.
Fig. 6.6: Changes in primary energy supply mix under CO₂ emissions reduction cases during 2013-2035.



Note: * Solar includes solar PV and solar thermal. Biomass excludes biofuels.

Fig. 6.7: Increase in renewable energy sources under CO₂ emissions reduction cases during 2013-2035.

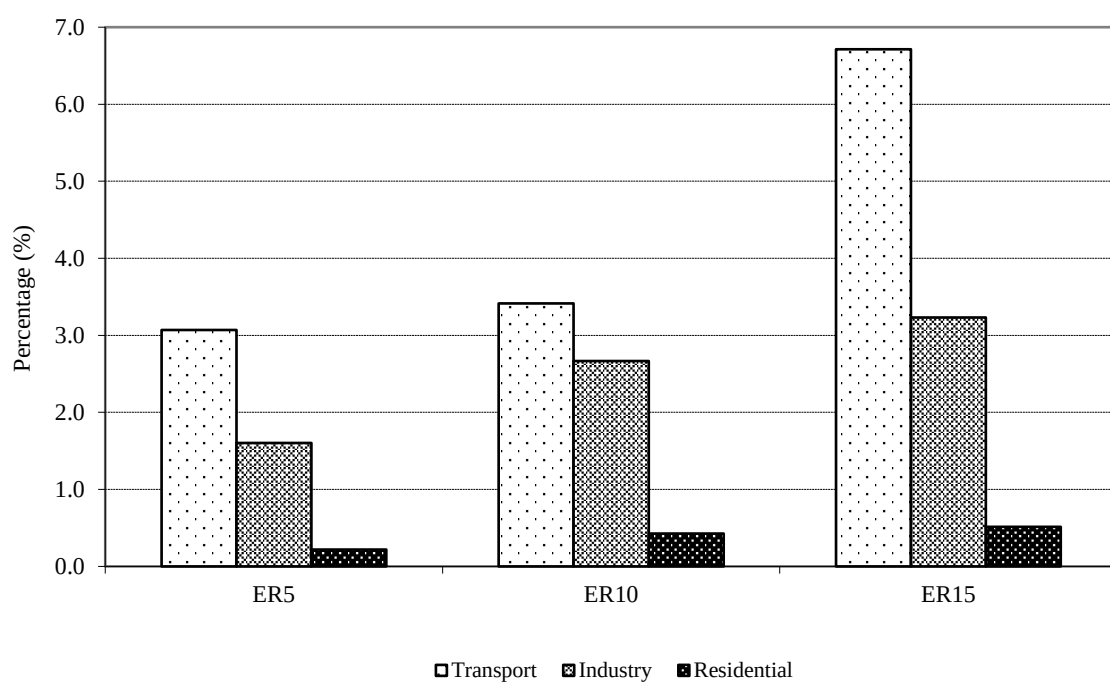


Fig. 6.8: Sector-wise reduction in final energy demand under CO₂ emission reduction target cases compared to base case during 2013-2035 (%).

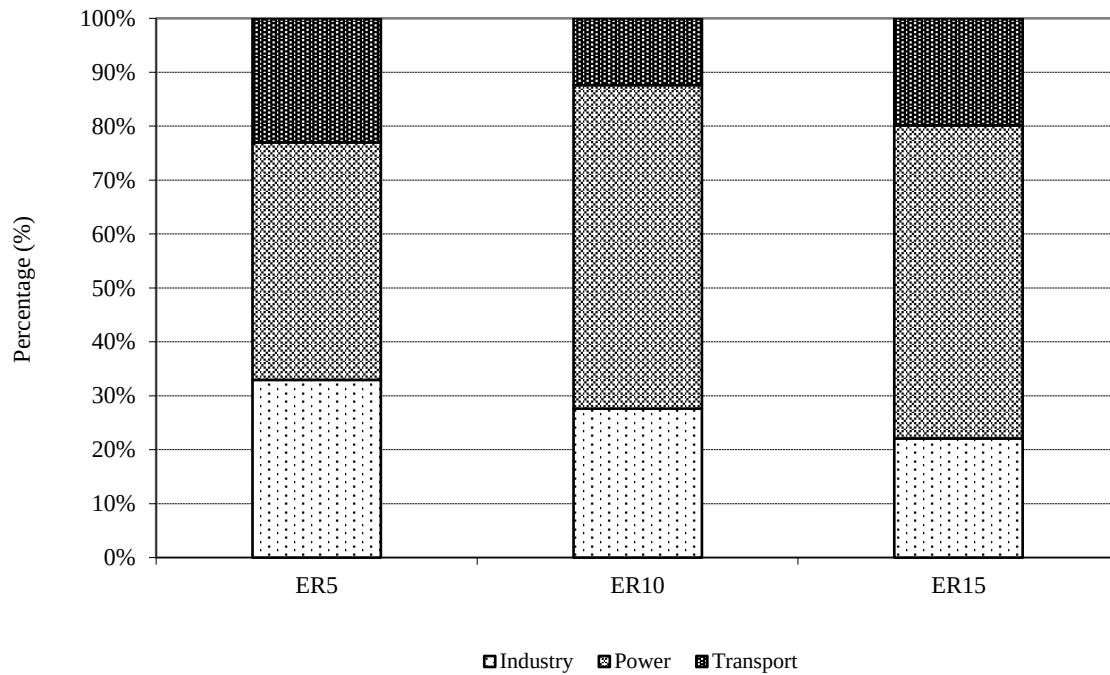
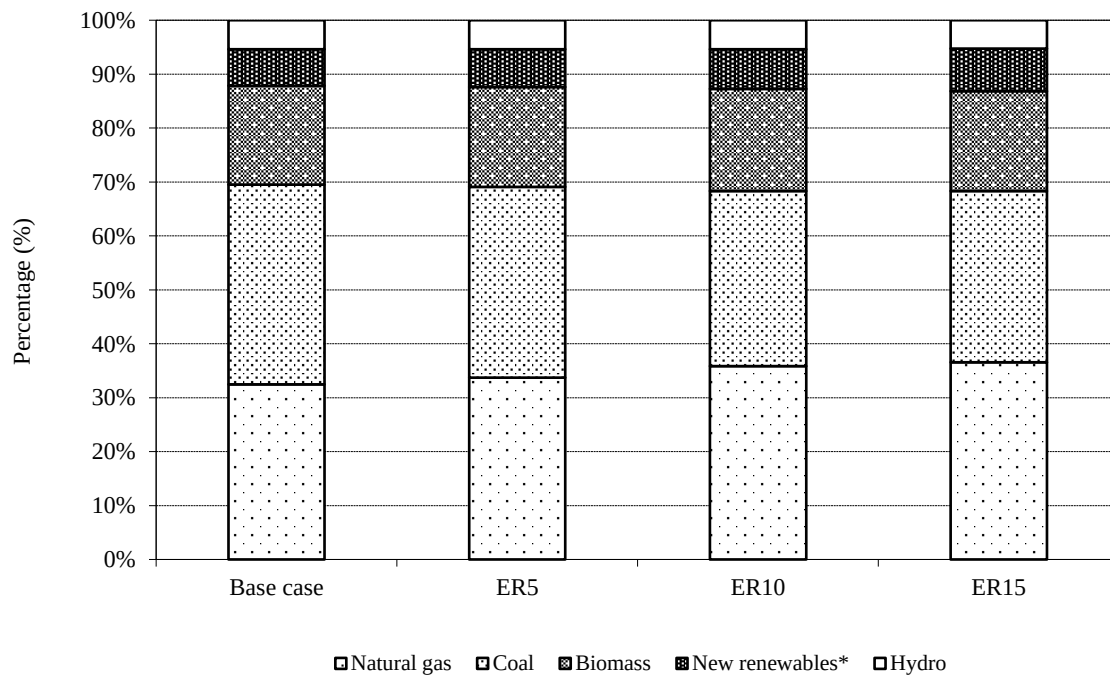


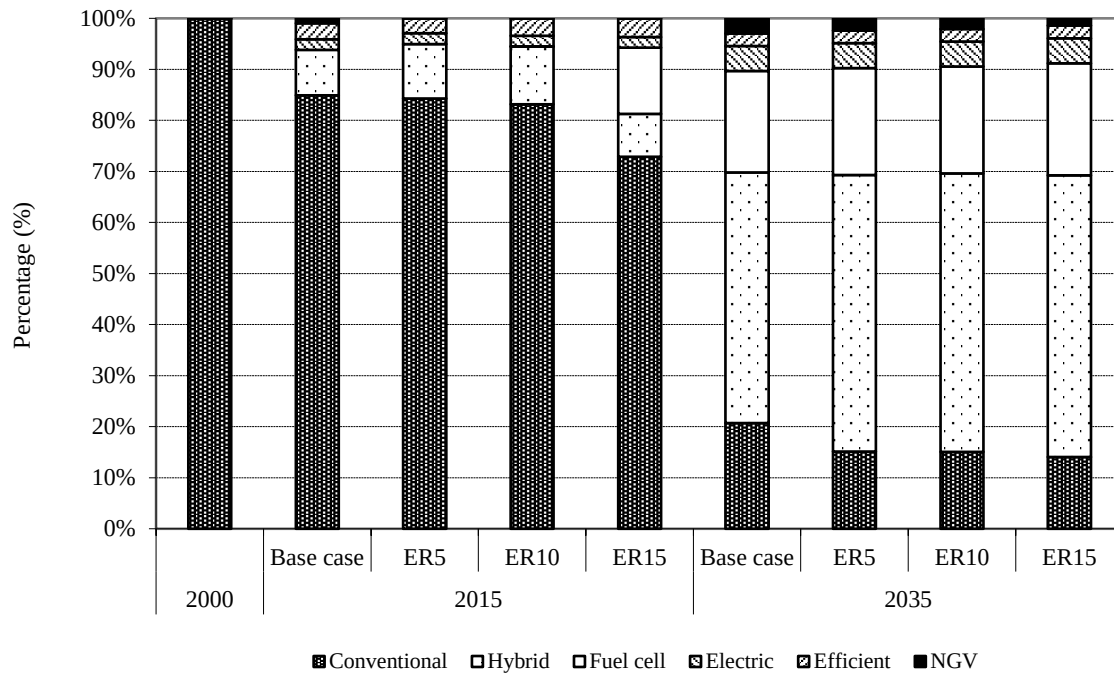
Fig.

6.9: Sector-wise contribution to CO₂ emissions reduction under ER cases during 2013-2035.



Note: New renewables include solar, wind, geothermal, biogas and MSW.

Fig. 6.10: Energy mix in electricity generation under the base case and ER target cases during 2013-2035.



Notes: Existing technologies include conventional gasoline, diesel, LPG and natural gas vehicle types. Efficient technologies include new versions of existing, hybrid vehicles, biodiesel direct injection and urban electric train.
Fig. 6.11: Technology penetration in road transportation under base case and ER target cases in 2000, 2015 and 2035.

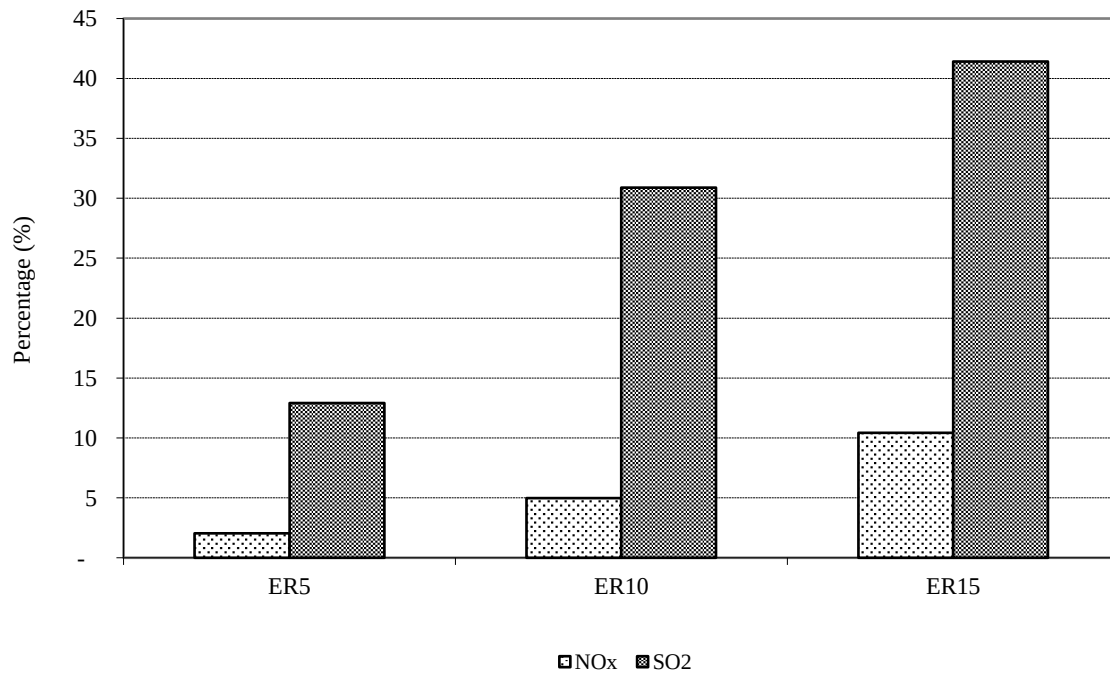


Fig. 6.12: Percentage reduction of SO₂ and NO_x emissions under ER target cases compared to base case during 2013-2035.

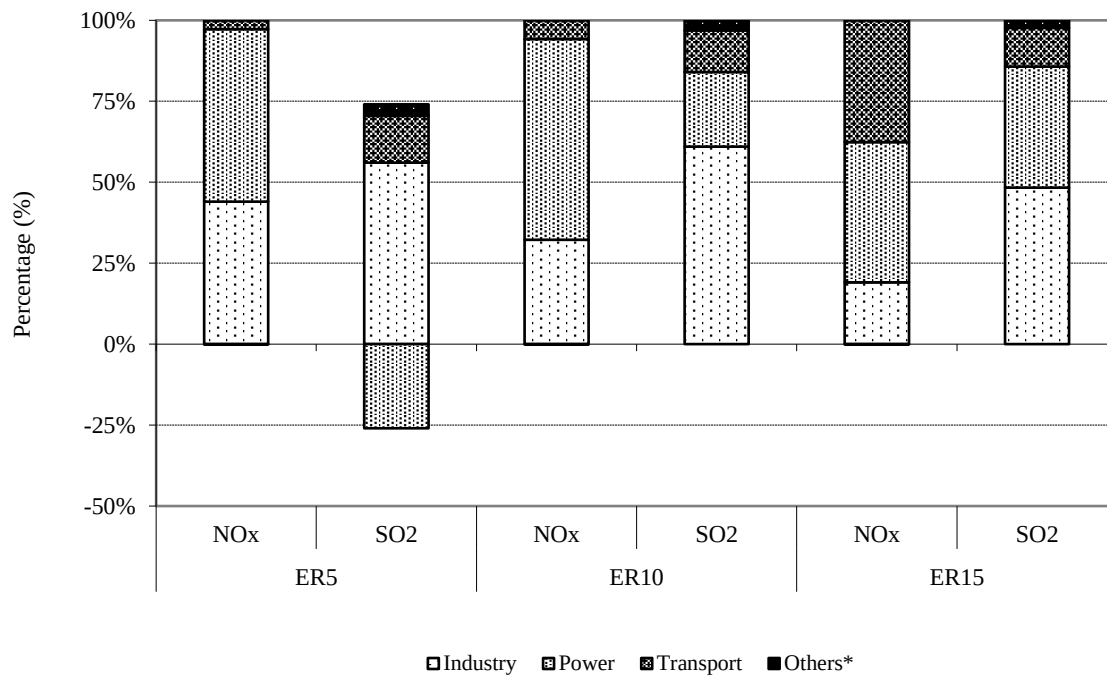


Fig. 6.13: Sector-wise percentage reduction of SO₂ and NO_x emissions under ER target cases compared to base case during 2013-2035.

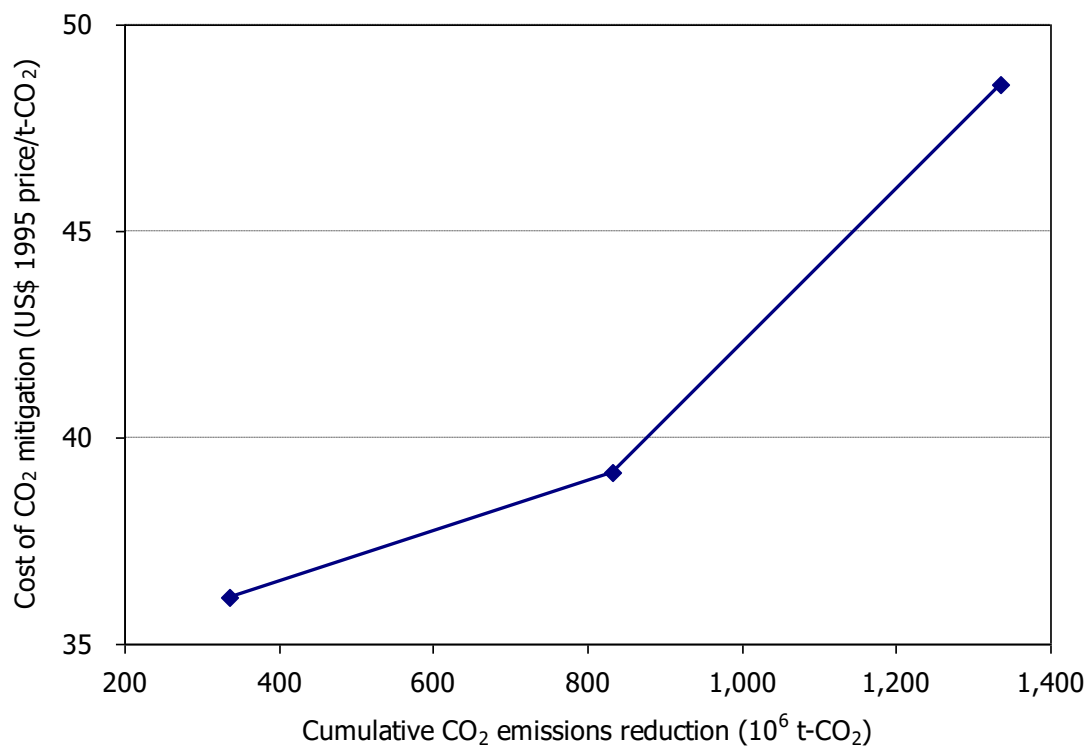


Fig. 12. Incremental cost of CO₂ emissions abatement during 2013-2035.

Table 6.1
Total electricity generation by plant type (TWh)

Technology type	Energy type	2000	2035			
			Base case	ER5	ER10	ER15
Fossil fuel and biomass based technologies						
Conventional steam	Lignite	16	20	20	21	-
	Natural gas	10	-	-	-	-
	Fuel oil	10	-	-	-	-
	Biomass	-	30	30	29	29
Integrated gasification combined cycle (IGCC)	Lignite	-	22	21	7	-
	Bituminous coal	-	35	34	35	33
Pressurized fluidized bed combustion (PFBC)	Lignite	-	12	10	-	-
	Bituminous coal	-	45	44	46	18
Combined cycle	Natural gas	39	-	-	-	-
	Fuel oil	-	-	-	-	-
Combined cycle-advanced	Natural gas	-	134	108	93	78
Gas turbine	Natural gas	-	-	-	-	-
	Fuel oil	-	-	-	-	-
Biomass integrated gasification combined cycle (BIGCC)	Biomass	-	39	39	40	40
Carbon capture and storage (CCS)		-	-	-	-	-
Integrated gasification combined cycle (IGCC)	Lignite	-	-	25	31	31
	Bituminous coal	-	30	30	31	31
Pressurized fluidized bed combustion (PFBC)	Lignite	-	-	2	17	31
	Bituminous coal	-	17	17	17	31
Advanced combined cycle	Natural gas	-	-	1	15	31
Renewables based technologies		-	-	-	-	-
Hydro		8	31	31	31	31
Wind		-	2	2	2	2
Solar PV		-	-	-	-	1
Solar thermal		-	-	1	12	11
Geothermal		-	6	6	6	6
MSW		-	6	6	6	6
Biogas		-	3	3	3	3
Co-generation	Coal	3	-	-	-	-
	Natural gas	6	-	-	-	-
Advanced technologies	Hydrogen	-	-	-	-	31
Total		93	433	431	445	445

Appendix 6.A

Classification of service demand component of energy system development of Thailand

Sector/sub sector	Energy Service	Number of Technology options	Activity Parameters
Agriculture	Pumping, tilling	4	Agricultural sector value added
Commercial	Air-conditioning, lighting, thermal use	9	
Industry			
Cement	Mixing, clinker production, milling, others	11	value added for cement
Steel	Primary steel production (coke, sinter, pig iron, crude and rolled steel production) Secondary steel production (crude steel and finished steel production)	44	Value added for steel
Sugar	Juice extraction, slurry production, crystallization and grinding	9	Value added for food
Paper	Pulp production, bleaching, stock production and others	11	Value added for paper
Others (chemicals, food and beverages, fabricated metals, textiles, others)	Thermal and mechanical energy	72	Value added of the respective sub sectors
Residential			
Metropolitan Electricity Authority (MEA)/Provincial Electricity Authority (PEA)	Air-conditioning, cooking, lighting, fan, iron, refrigerator, television, other	32	Number of house holds, appliance ownership per household
Transport			
Passenger	Road (passenger car, micro bus and pickup, van and pickup, three wheeler, urban taxi, fixed route taxi, motor tricycle taxi, motorcycle, fixed route bus, bus for hire, private bus, small rural bus, urban train)	79	Per capita GDP, average distance traveled by each vehicle, occupancy factor #
	Rail and air	9	Per capita GDP
Freight	Road and water	12	Transport sector value added

Note: *Appliance ownership is the number of houses that own a specific appliance as a fraction of total number of houses. # Occupancy factor is the average number of persons per vehicle at a given time of travel.

Appendix 6.B

List of sources used in Thai Energy system development

Category	Source
Economic	NESDB, 2003; NIDA, 2006
Demographic	UN, 2004
Service demand	KMIT, 1997; KMUTT, 2003a,b; NEPO, 1999; NSO, 2003, IEA, 2005
Technology type	Kainuma et al., 2003; NEPO, 1999; Tseng et al., 2005;
Energy	DEDE, 2006a,b,c; DEDE, 2004; EGAT, 2004; Greacen, 2005; IEA, 2004b; Prasertsan and Sajjakulnukit, 2006; Sajjakulnukit and Verapong, 2003; Santisirisomboon et al., 2003
Emissions factors	IPCC, 1996; Foell et al., 1995
Appliance stock in the start year	NEPO, 1999; KMUTT, 2003a,b; DLT, 2000

Appendix 6.C

Domestic fossil fuels resource availability in Thailand (Mtoe)

Fuel type	2005	2015	2033
Coal	8.4	11.5	15.2
Crude oil	0.4	0.5	0.6
Natural gas	14.6	19.2	10.2

Source: NEPO (1999).